

Blockchain Practice – Bitcoin

Ex1: A Bitcoin block contains 11 total transactions, which includes the mandatory Coinbase transaction.

How many leaf nodes will the Merkle tree have? Explain how Bitcoin handles the odd transaction.

Ex2: If a block were to contain 16 transactions, how many hashes are required in a Merkle proof to verify that one specific transaction is included in the block?

Ex3: Alice has a single UTXO worth 2.5 BTC. She sends 2.0 BTC to Bob and creates a change output for herself worth 0.4 BTC.

- a. Is this transaction valid? Explain using the Input/Output rule.
- b. What is the transaction fee for the miner?

Ex4: A miner is attempting to find a valid block hash. The current Difficulty Target is 00000000ABCD.... The miner calculates a Block Hash of 00000000BCDE....

- a. Is this block valid according to Bitcoin's rules? Explain.
- b. List three fields in the block header that cannot be changed by the miner during the re-mining process.